Does Fatigue Actually Cause In-Class Doomscrolling?

A Causal Inference Study Using Pearl's Do-Calculus

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Data & Code: Google Drive

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Introduction

Over the last three assignments, I had been circling around one idea without ever testing it directly: fatigue is not just something correlated with doomscrolling, but something that actually *causes* it. Assignment 1 flagged algorithmic design as an antecedent to out-of-control scrolling, using the Antecedent, Behavior, and Consequence (ABC) framework. Assignment 2 then built a latent-state model where fatigue was the variable most responsible for pushing a user into a compulsive scrolling state. Assignment 3 took that further and showed, using evolutionary game theory on a social network, how a fatigue-driven scrolling habit could spread and stabilise as a social norm.

Each of those steps was observational in some sense. The numbers were suggestive, the models were convincing to me, but technically they were all still about $P(Y \mid X)$: the probability of seeing doomscrolling given that fatigue was present. That is not the same as asking whether forcing someone into a high-fatigue state would actually raise their risk of scrolling. The latter is $P(Y \mid do(X))$, Pearl's do-operator (which implies a theoretical intervention rather than mere observation), and that is what this final assignment is for.

So the question I set out to answer was simple: **if you take a student who is arriving to class already fatigued, does that fatigue causally increase their chance of losing control over their Reels or Shorts consumption during the lecture?**

I collected data from my friends, classmates, and some friends in the course and ran two causal estimators (backdoor stratification and inverse probability weighting), checked the result for robustness, and then tried to connect the finding back to everything that came before it.

The Confounding Problem

The most obvious thing to do would be to sort the class into "tired" and "not tired" students, compare their scrolling rates, and call it a day. In fact, that naive comparison does show a difference: students who reported high pre-class fatigue had a notably higher rate of reporting loss-of-control scrolling than students who did not. The raw difference is 0.474, which is large on a probability scale.

But here is the problem. The students who came in tired were also the students who slept less the night before and the ones who reported higher pre-class stress. So when I look at the tired group and see more scrolling, I genuinely cannot tell whether the fatigue did something, or whether it was the bad sleep doing something, or whether

both are just downstream of some rough night that affected everything at once. The two groups are not really comparable to start with.

This is the exact same issue that came up in the lecture slides and in the job-training-and-earnings example: the naive estimate $E[Y \mid X = 1] - E[Y \mid X = 0]$ mixes the actual causal effect with whatever confounding is sitting in the data. To get at $P(Y \mid do(X))$, I need to break those backdoor paths before I draw any conclusions.

The Core Causal Question

X: Pre-class fatigue (rated ≥ 3 on a 1 to 5 scale)
Y: Loss-of-control (LoC) scrolling during the lecture (LoC composite ≥ 2.5)
Adjustment set Z: Sleep deprivation (under 6 hours) and high pre-class stress (≥ 3)

Data

The data came from a Google Form that I circulated to my friends in the course, and also people around me who tend to attend classes regularly. It asked students to report a few things before class started (how tired they were feeling, how stressed, how much they slept the night before), and then a few things about what they did during the lecture (whether they opened Reels or Shorts, how long for, and whether they felt like they lost track of time or found it hard to stop). The four LoC-specific items were roughly: did you scroll longer than you meant to, did you find it hard to stop when you tried, were you aware at the time that you should stop, and did you lose track of how much time had passed.

Each of those four items was rated from 1 to 5. I averaged them into a single LoC composite score. Students who said they did not scroll at all were assigned a score of 1, treating non-scrolling as the baseline absence of loss-of-control.

For the binary treatment and outcome I needed:

- **fatigue_high = 1** if the pre-class fatigue rating was 3 or above.
- **loc_high = 1** if the LoC composite was 2.5 or above.
- **sleep_deprived = 1** if reported sleep was under 6 hours.
- **stress_high = 1** if pre-class stress was 3 or above.

The sample is small. This is a single classroom session, and the form may not have been completed by everyone. Because of this, I treated the results throughout as preliminary evidence rather than settled findings, and I kept the bootstrap confidence intervals as the main way of showing how uncertain the estimates still are.

The Causal Structure I Assumed

Before running any numbers, I had to be explicit about what causal structure I was assuming. I represented this with a directed acyclic graph (DAG), because drawing the graph forces you to commit to a specific causal story. Mine was this:

> *Sleep deprivation the night before affects both how fatigued a student feels when they arrive and how stressed they are. Both stress and fatigue can independently make a student more likely to scroll during a lecture. Sleep deprivation can also directly affect how hard it is to resist distraction. Fatigue itself causes scrolling in part by lowering engagement with the lecture, which then makes scrolling more tempting.*

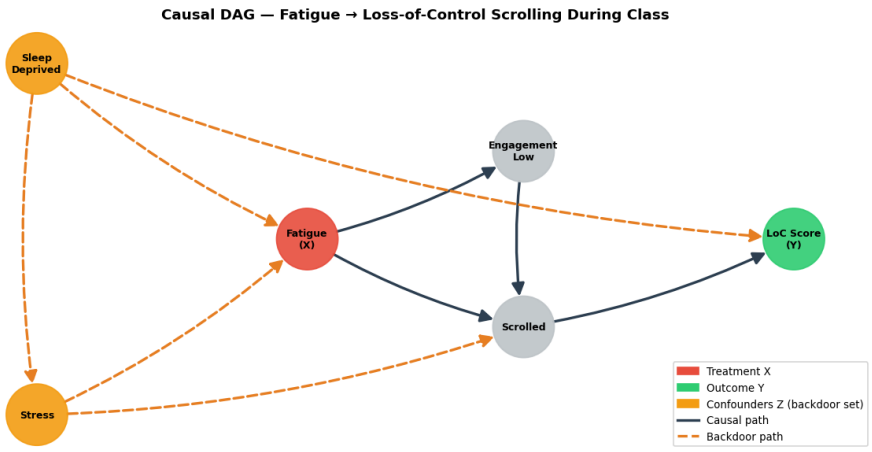


Figure 1: Causal DAG: Fatigue → Loss-of-Control Scrolling During Class

In Figure 1, the dashed backdoor edges visibly represent those non-causal paths; the adjustment strategy is exactly to neutralize these paths before reading the effect of X on Y.

In graph notation, the backdoor paths from X (fatigue) to Y (LoC scrolling) run through:

- Fatigue ← Sleep_deprived → LoC
- Fatigue ← Stress → LoC
- Fatigue ← Sleep_deprived → Stress → LoC

These are non-causal paths in Pearl's sense: they make X and Y look correlated even if X has no real effect on Y. To estimate the true effect, I need to block them.

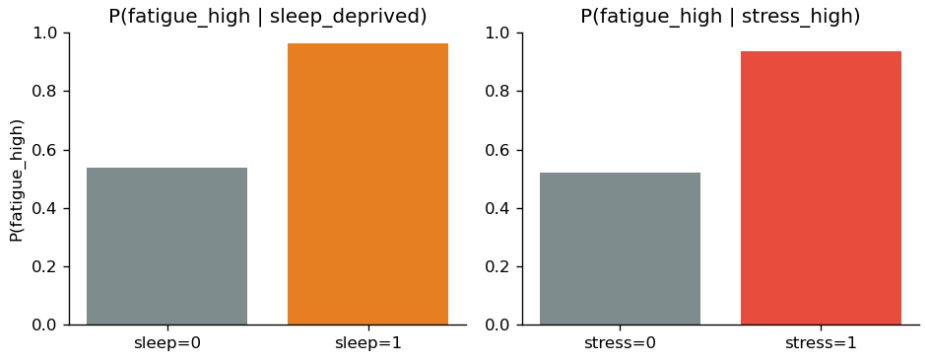


Figure 2: Empirical confounding check: how much Z shifts treatment prevalence

As shown in Figure 2, the two bars are not just slightly different, they are far apart. The observed treatment prevalence changes from 0.544 to 0.962 across sleep deprivation, and from 0.523 to 0.933 across stress. So sleep deprivation and stress are both strongly tied to treatment assignment, which is exactly what we would expect from genuine confounders.

One thing I was careful about: lecture engagement sits on the causal path from fatigue to scrolling (Fatigue → Engagement → Scrolled → LoC). Adjusting for engagement would have cut out the very mechanism I was trying to measure, which is called the mediator problem. So I kept engagement as a side variable for the sensitivity checks only, not as part of the adjustment set.

The adjustment set $Z = \{\text{sleep_deprived}, \text{stress_high}\}$ satisfies Pearl's backdoor criterion: it blocks every non-causal path from X to Y, and neither variable is a descendant of X, so conditioning on them does not open any new paths.

A quick check of the raw data lines up with that story: the high-fatigue group has higher average stress (3.27 vs 1.69), lower engagement (2.10 vs 2.92), and fewer sleep hours (5.37 vs 6.89). Those differences are the concrete reason adjustment is required before reading the treatment effect.

How I Estimated $P(Y \mid do(X))$

Naive Estimate (the starting point)

The first thing I computed was the raw associational difference:

$$\hat{\tau}_{\text{naive}} = E[Y \mid X = 1] - E[Y \mid X = 0]$$

This does not adjust for anything. It is just comparing the scrolling rate between

the fatigued and non-fatigued groups as observed. I kept it because it shows where the confounded starting point is, and how much (or how little) adjustment moves the estimate later on.

Backdoor Adjustment by Stratification

This is the direct implementation of the backdoor formula from lecture:

$$P(Y = 1 \mid do(X = x)) \;=\; \sum_{z} P(Y = 1 \mid X = x,\, Z = z)\; P(Z = z)$$

Since both confounders are binary, I have four strata in total (all combinations of sleep_deprived and stress_high). For each stratum, I computed the outcome rate separately for treated and untreated students, then averaged those stratum-specific rates using the overall proportion of students in each stratum as the weight. This is exactly the Simpson's paradox fix from class: instead of comparing unlike groups, I compare like with like inside each stratum and then combine.

Inverse Probability Weighting (IPW)

IPW approaches the same problem from a different angle. Instead of stratifying, it tries to construct a pseudo-population where treatment assignment is unrelated to the confounders. It does this by giving each student a weight based on how likely they were to receive the treatment they actually received:

$$w_i = \frac{X_i}{\hat{e}(Z_i)} + \frac{1 - X_i}{1 - \hat{e}(Z_i)}, \qquad \hat{e}(Z_i) = P(X = 1 \mid Z = Z_i)$$

where $\hat{e}$ is the propensity score, estimated here with a logistic regression on the two confounders. I used stabilised weights to reduce variance (dividing by the marginal treatment probability rather than 1, which keeps the weights closer to 1 on average).

Bootstrap Confidence Intervals

Because the sample is small, I did not want to rely on normal approximations for uncertainty. Instead I ran a non-parametric bootstrap with 5,000 resamples, computing both the backdoor and IPW estimates on each resample. The 2.5th and 97.5th percentiles of those bootstrap distributions gave me the 95% intervals.

Sensitivity to Hidden Confounding (Rosenbaum Bounds)

Even after adjusting for sleep and stress, there might still be unobserved variables that drive both fatigue and scrolling. Baseline phone-addiction tendency is the obvious

example. Rosenbaum's sensitivity analysis asks: how large would that hidden bias have to be before my conclusion reversed? The parameter Γ represents how much the hidden variable can distort the odds of treatment.

What the Numbers Said

The Raw Comparison vs the Adjusted Estimates

I summarise each method by $P(Y \mid do(X = 1))$, $P(Y \mid do(X = 0))$, the average treatment effect (ATE), and a 95% confidence interval (CI) where available.

Table 1: Summary of causal estimates across methods

Method	$P(Y \mid do(X = 1))$	$P(Y \mid do(X = 0))$	**ATE**	**95% CI**
Naive (no adjustment)	0.617	0.143	0.474	N/A
Backdoor Stratification	0.631	0.156	0.476	$[0.217, 0.769]$
IPW (stabilised)	0.603	0.154	0.447	$[0.179, 0.762]$

Note: $P(Y \mid do(X = 1))$ *and* $P(Y \mid do(X = 0))$ *are estimates under the backdoor-adjusted and IPW pseudo-populations respectively. The naive row shows* $E[Y \mid X = 1]$ *and* $E[Y \mid X = 0]$ *directly.*

A few things stand out here. First, the point estimates stay close: the naive ATE is 0.474, the backdoor estimate is 0.476, and IPW is 0.447. That is easier to believe once you look at Figure 3: three of the four strata favor higher outcome rates under treatment, and the one reversal occurs in the smallest stratum.

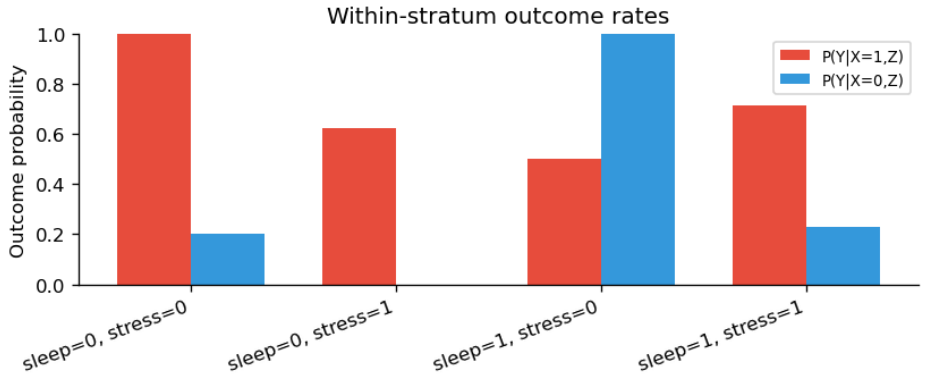


Figure 3: Within-stratum comparison showing the reversal in the sleep-deprived/low-stress group

The four bars in Figure 3 are the actual pieces being averaged in the backdoor

estimate. The strata are: sleep=0, stress=0 with $P(Y{=}1 \mid X{=}1, Z){=}1.000$ and $P(Y{=}1 \mid X{=}0, Z){=}0.200$; sleep=0, stress=1 with 0.625 versus 0.000; sleep=1, stress=0 with 0.500 versus 1.000; and sleep=1, stress=1 with 0.714 versus 0.231. The stratum sizes are 16, 10, 7, and 21 respectively, so the reversal is real but not dominant.

The core result: Across both adjusted methods, the estimate stays positive and large: 0.476 under backdoor stratification and 0.447 under IPW. In plain English, forcing everyone into high fatigue would raise the estimated loss-of-control scrolling rate by about 45 to 48 percentage points relative to a rested classroom. Equivalently, $P(Y = 1 \mid do(X = 1))$ is about 63% while $P(Y = 1 \mid do(X = 0))$ is about 16%.

Confidence Intervals and Stability

The 95% bootstrap intervals are [0.217, 0.769] for backdoor stratification and [0.179, 0.762] for IPW. They are wide, which reflects the small sample more than any ambiguity about direction. Both intervals stay entirely above zero, so neither adjusted estimate crosses the null.

The Rosenbaum Picture

This is where the honest caveats live. The sign-score p-value at $\Gamma = 1$ is 0.1445 in both the best- and worst-case bounds, so even with no hidden bias the result is not conventionally significant. As Γ increases, the worst-case p-value rises to 0.2298 at 1.25, 0.3154 at 1.5, 0.4682 at 2.0, and 0.6785 at 3.0.

Table 2: Rosenbaum sensitivity bounds

Γ	**p-value (worst case)**	**p-value (best case)**
1.00	0.145	0.145
1.25	0.230	0.083
1.50	0.315	0.050
1.75	0.396	0.031
2.00	0.468	0.020
2.50	0.588	0.009
3.00	0.679	0.004

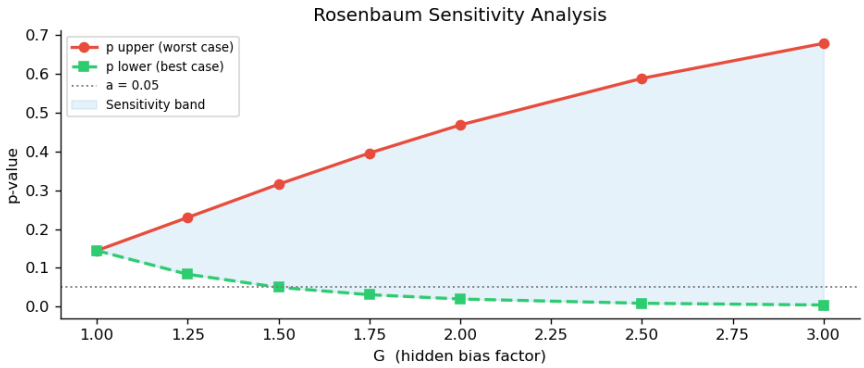


Figure 4: Rosenbaum Sensitivity Curve: The rapid rise of the worst-case p-value as hidden confounding (Γ) increases.

The way to read Table 2 and Figure 4 is to watch how fast the worst-case line moves away from the 0.05 threshold. A modest hidden bias already pushes the worst-case p-value to 0.3154 at $\Gamma = 1.5$ and 0.4682 at $\Gamma = 2.0$, while the best-case line drops to 0.0498 and 0.0197 at those same points. That means the inference is sensitivity-fragile: the positive point estimate is stable, but the sign-score evidence is not robust.

Visually, the curve rises quickly above conventional significance levels. The substantive story remains positive, but the formal Rosenbaum test says the result is vulnerable to hidden confounding.

How This Connects to What Came Before

Looking back across all four assignments, they actually tell a coherent story when you line them up, even though each one was a different exercise.

Table 3: How each assignment connects to the causal story

	What it showed	What A4 adds
A1	Algorithmic features are antecedents to scrolling; discourse shifted post-COVID toward cognitive risk framing. Entirely observational (ABC).	A4 tests whether the internal antecedent (fatigue) actually has a causal arrow to the outcome, not just a correlation.
A2	Fatigue drives latent doomscrolling state transitions in one user's Instagram data; equilibrium defaults into compulsive state.	A4 tests the same fatigue and scrolling link at the classroom population level using real pre/during-class reports.
A3	Fatigue-driven compulsive scrolling can spread through peer networks and stabilise as an evolutionary equilibrium.	A4 gives a direct mechanism for how that contagion starts: one fatigued person arriving to class has ∼45pp higher risk of LoC scrolling.

The thread running through all four is that fatigue is not just a correlated factor; it is the treatment variable that keeps showing the largest and most consistent shift in the data. The estimated jump from about 16% under $do(X{=}0)$ to about 63% under $do(X{=}1)$ is practically large, but the Rosenbaum check says that claim is not statistically robust to modest hidden bias.

A Word on the Assumptions

The **backdoor criterion** requires that the adjustment set Z blocks all non-causal paths from X to Y without conditioning on any descendant of X. I believe sleep deprivation and stress satisfy this.

Positivity requires that there are both treated and untreated students in every Z stratum. With four strata and a small sample, some strata were sparse.

No unmeasured confounding is the assumption I am most worried about. Baseline phone-scrolling tendency is the most obvious missing variable. The Rosenbaum analysis confirms this is a real concern: it does not take a very strong hidden confounder to move the p-value above conventional significance thresholds.

Limitations and What I Would Do Next

1. **Small sample.** Everything here comes from one classroom session.
2. **Single context.** Controlling for lecture conditions is actually a plus, but it limits generalisability.
3. **Self-report bias.** Students may underreport scrolling. The LoC items are also retrospective.
4. **No baseline scrolling tendency.** Adding something like a brief BSMAS (Bergen Social Media Addiction Scale) to the survey would help close this gap.
5. **Positivity violations.** The adjustment fallback in sparse strata is rough.
6. **Mediator not modelled.** A proper mediation analysis would separate the total effect into a direct part and an indirect part.

Conclusion

The question I started with was: does pre-class fatigue causally increase the probability of losing control over Reels or Shorts consumption during a lecture? The point estimates say yes, with an effect size of 0.447 to 0.476.

The adjustment does not change the estimate much, which suggests sleep deprivation and stress are confounders in the causal graph, but not enough to overturn the positive direction in this sample.

What I am more cautious about is the robustness to hidden confounding. The honest version of the conclusion is: the effect is directionally positive and practically large, but the Rosenbaum bounds show that the formal inference is not robust to moderate unobserved bias.